A characterization of the top-trading-cycles mechanism via endowment-position strategy-proofness

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Abstract

Strategy-proofness is central to axiomatic characterizations of the top-trading-cycles (TTC) mechanism, but it constrains every possible misreport of a complete preference order. We ask how much incentive compatibility is actually needed to identify TTC in the Shapley–Scarf housing market. We hold fixed an agent's ranking of all potential trade targets and vary only the position of her endowed house, which is her reported acceptability cutoff between trading and staying. Endowment-position strategy-proofness requires that no repositioning of this cutoff be profitable. It imposes no incentive restriction on deviations that rerank trade targets. Our main result is that *pair-efficiency*, *respecting-improvement*, and *endowment-position strategy-proofness* characterize TTC. A local-to-global observation shows that *endowment-position strategy-proofness* can be verified using only the one or two adjacent swaps that move the endowment. Thus, once pair-efficiency and respecting-improvement are imposed, this one-dimensional incentive compatibility identifies TTC and thereby recovers full and even group strategy-proofness. Endowment-position strategy-proofness disciplines behavior within each endowment-position fiber, whereas respecting-improvement connects the fibers. Replacing the latter by *individual-rationality* preserves the characterization at $n = 3$ but makes it fail for every $n \geq 4$; natural weakenings of the restricted incentive requirement also fail.

Keywords: housing markets, top-trading-cycles, pair-efficiency, respecting-improvement, strategy-proofness, truncation
JEL: C78, D47, D71

1. Introduction

In a Shapley–Scarf housing market (Shapley and Scarf, 1974), a group of agents each owns one indivisible object, each wants exactly one object, and money is unavailable. The designer's problem is to reallocate. The top-trading-cycles (TTC) mechanism, attributed by Shapley and Scarf (1974) to David Gale, selects the unique strong core allocation of every such market (Roth and Postlewaite, 1977), and it is strategy-proof (Roth, 1982) and even group strategy-proof (Bird, 1984; Sandholtz and Tai, 2024). Its prominence in market design rests on this combination of properties, and therefore on the axiomatic characterizations that pin down exactly which combinations are decisive.

Strategy-proofness gives this theory its incentive foundation, but it does so by imposing global discipline: it compares truthful reporting with *every* alternative complete ranking. In an endowed market, however, a report carries two conceptually different

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kinds of information. It ranks the houses an agent might trade for, and it locates her own house among them. The latter determines which trades she declares preferable to keeping what she owns. We ask whether TTC can be identified by disciplining only this second part of the report, without directly restricting manipulations that reorder potential trade targets. The reporting domain remains unrestricted; what is restricted is the set of deviations required to be unprofitable. In this sense, we decompose strategy-proofness and ask which comparisons do the identifying work once efficiency and ownership response are imposed.

Fix an agent and hold fixed how she ranks all the houses she does not own. Her only remaining choice is where to insert her endowed house. Consider a homeowner who regards two alternative houses as better than staying and a third as worse. She may instead report that only her favorite alternative is worth moving for, without changing how she ranks the alternatives themselves. Her endowed house has crossed just one neighboring alternative; only the line between trading and staying has moved. The position of her endowed house is thus her reported acceptability cutoff: how good an alternative must be before she is willing to move. Raising the endowed house in her report is the classical truncation manipulation — claiming to be harder to please; lowering it overstates willingness to trade. We call a mechanism *endowment-position strategy-proof* if no such repositioning, upward or downward, ever pays. Formally, such a report change is an *endowment repositioning*: only the endowment's position may differ. Equivalently, this is the bidirectional form of endowment-truncation-proofness. Throughout, this is the formal primitive and the acceptability cutoff its economic interpretation. Thus, *endowment-position strategy-proofness* imposes incentive compatibility on that cutoff and nowhere else in the preference order.

Paired kidney exchange gives this cutoff a concrete interpretation. In models where observable clinical characteristics support a common ranking of compatible kidneys, a pair's willingness to participate can be represented by the minimum acceptable improvement over its intended donor. Holding that ranking fixed while changing the threshold moves the intended donor's kidney through the list. Quality-based rankings are studied by Nicolò and Rodríguez-Álvarez (2012, 2017), and compatible-pair participation by Chipman et al. (2022). This application motivates a distinguished reporting margin rather than a restricted preference domain: the mechanism remains defined for heterogeneous rankings while the incentive requirement singles out changes in the participation cutoff.

Endowment-position strategy-proofness has an exact local test: it is enough to rule out gains from moving the endowment across one neighboring object. Respecting-improvement has an analogous adjacent test, while pair-efficiency is pairwise.

Our main result is that the TTC mechanism is the unique mechanism satisfying *pair-efficiency*, *respecting-improvement*, and *endowment-position strategy-proofness* (Theorem 1). Equivalently, once pair-efficiency and respecting-improvement are imposed, incentive compatibility along the single endowment-position coordinate is amplified into full and even group strategy-proofness. Initial ownership therefore selects the economically distinguished coordinate that carries the identifying power.

Three further results map the boundary of the characterization.

First, Theorem 2 isolates *respecting-improvement*'s cross-fiber role. The combination of *individual-rationality*, *pair-efficiency* and *endowment-position strategy-proofness* characterizes TTC at $n = 3$, whereas for every $n \geq 4$ a family of mechanisms satisfies

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all three and differs from TTC. Endowment-position strategy-proofness constrains a mechanism within each set of profiles sharing a fixed ranking of non-owned objects. From four agents onward, a nested family of longer trading cycles can be coordinated inside one such set without being detected by pair-efficiency. Respecting-improvement links these sets and supplies the missing cross-fiber discipline.

Second, the upward half of *endowment-position strategy-proofness* is insufficient at this efficiency level. Replacing *endowment-position strategy-proofness* by the one-sided *endowment-truncation-proofness* fails already with three agents: the mechanism β satisfies *pair-efficiency*, *respecting-improvement* and *endowment-truncation-proofness* but differs from TTC (Proposition 2).

Third, truncation-invariance provides a different local benchmark. It imposes invariance only at the report declaring exactly the assigned object and better to be acceptable. With four agents there is a mechanism that is individually-rational, Pareto-efficient, respects improvement and is truncation-invariant, yet differs from TTC (Proposition 3).

The proof studies the first TTC cycle that a deviating mechanism fails to execute. Respecting-improvement pins owners while outsiders' reports are edited, and a common-sink construction rules out assignments from earlier TTC cycles. The failed cycle can then be separated and used to induce a smaller counterexample, which strong induction excludes.

The rest of the paper is organized as follows. Section 2 relates the paper to the literature. Section 3 presents the model and the axioms. Section 4 states the results and gives the shorter proofs. Section 5 proves the main characterization. Section 6 concludes.

2. Related literature

Klaus et al. (2025) characterize TTC by *pair-efficiency*, *respecting-improvement* and full *strategy-proofness*. This is the closest benchmark for Theorem 1. We keep their efficiency and ownership-response requirements unchanged and replace full *strategy-proofness* by *endowment-position strategy-proofness*, which restricts incentive compatibility to the endowed-house cutoff. Their Lemma 1 shows that *pair-efficiency* and *respecting-improvement* imply *individual-rationality*. Their Example 3 supplies the mechanism β used to show that *strategy-proofness* cannot simply be dropped; Proposition 2 establishes the additional fact that β satisfies one-sided *endowment-truncation-proofness*.

Locality here has two meanings. The first concerns the domain of deviations: *endowment-position strategy-proofness* checks only the one-dimensional family obtained by repositioning the agent's own endowment. The second is combinatorial locality: by Lemma 1, adjacent constraints generate every comparison on that fiber. Full strategy-proofness itself has an adjacent-swap test on the unrestricted strict domain (Carroll, 2012; Sato, 2013), but it uses all $n-1$ adjacent swaps of a reported ranking; *endowment-position strategy-proofness* retains only the one or two involving the endowment. Respecting-improvement likewise has an exact adjacent test, now in one other agent's report, and pair-efficiency is pairwise. Related analyses of local incentive constraints for ordinal mechanisms include Cho (2016) and the type-graph framework of Kumar et al. (2021).

Goel and Tamura (2026) take a complementary route. They retain full *strategy-proofness* but restrict the preference domain, and show that a top-two richness condition

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extends the *individual-rationality*, *pair-efficiency* and *strategy-proofness* characterization of TTC. Their proof uses auxiliary reports that place an agent's endowment immediately below her best remaining object. We keep the preference domain unrestricted and instead restrict the deviations that must be unprofitable.

Ma (1994) characterizes TTC by *individual-rationality*, *Pareto-efficiency* and *strategy-proofness*; short proofs are given by Svensson (1999), Anno (2015) and Sethuraman (2016). Ekici (2024) weakens *Pareto-efficiency* to *pair-efficiency*, which forbids only mutually beneficial two-agent swaps, and shows that *individual-rationality*, *pair-efficiency* and *strategy-proofness* still characterize TTC; Ekici and Sethuraman (2024) give a short proof. Several papers instead replace *strategy-proofness* by conditions that restrict smaller classes of deviations. Fujinaka and Wakayama (2018) use endowments-swapping-proofness, which constrains manipulation of endowments rather than of preferences. Chen and Zhao (2021) use rank monotonicity, a monotonic response of the selected allocation to truncations at that allocation. Altuntaş et al. (2023) use drop strategy-proofness, which rules out a gain from demoting one unowned object to the bottom of the report. A drop may reorder potential trade targets and therefore need not stay within an endowment-position fiber. Chen et al. (2025) give two characterizations using upper invariance: one combines it with *individual-rationality* and *pair-efficiency*, and the other with *individual-rationality* and endowments-swapping-proofness. Upper invariance holds an assignment fixed when objects below it are reordered and, under *individual-rationality*, implies *endowment-position strategy-proofness* strictly (Proposition 1). The four-agent member of Appendix A shows that *individual-rationality*, *pair-efficiency* and *endowment-position strategy-proofness* do not imply upper invariance; by contrast, Theorem 1 restores TTC when *respecting-improvement* replaces *individual-rationality*. This comparison locates the cross-fiber role of *respecting-improvement* relative to upper invariance.

Within this literature, truncation strategies are a familiar class of manipulations in matching markets (Roth and Rothblum, 1999; Ehlers, 2008), and *endowment-truncation-proofness* asks that shading one's endowment upward never pay. The condition is one-sided by construction. It is nonetheless strong enough in some combinations: Coreno and Feng (2026, Theorem 5) obtain a characterization by *individual-rationality*, *Pareto-efficiency*, and truncation-proofness as the single-object specialization of their multi-object model. Proposition 3 supplies the complementary boundary at the same efficiency level: even after respecting-improvement is added, truncation-invariance cannot replace truncation-proofness. Coreno and Feng (2025) replace *Pareto-efficiency* by endowments-swapping-proofness and show that this axiom cannot in turn be replaced by *pair-efficiency*: *individual-rationality*, *pair-efficiency* and truncation-proofness are insufficient. Proposition 2 strengthens this boundary in the present environment. Since *pair-efficiency* and *respecting-improvement* imply *individual-rationality*, the mechanism β shows that the insufficiency persists even after *respecting-improvement* is added. Biró et al. (2024) introduce *respecting-improvement* in the Shapley–Scarf market, motivated by kidney exchange — a patient should not be penalized for bringing a better donor — and show that TTC respects improvement; Schlotter et al. (2025) and Ehlers (2026) extend the analysis, the latter showing that *individual-rationality*, *strategy-proofness* and non-bossiness together imply *respecting-improvement*.

Beyond the Shapley–Scarf model, TTC and its extensions have been characterized in house allocation with existing tenants and in priority-based assignment. Pápai

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(2000) and Pycia and Ünver (2017) characterize hierarchical exchange and trading-cycles rules using *Pareto-efficiency*, *strategy-proofness* and non-bossiness-type conditions. Abdulkadiroğlu and Sönmez (2003) bring TTC to school choice; Morrill (2013) and Chen et al. (2023) give axiomatizations there. Morrill and Roth (2024) survey the area.

3. The model

3.1. Environment and TTC

There is a finite set of **agents** $N = \{1, 2, \ldots, n\}$ with $n \geq 3$. Each agent $i \in N$ is endowed with exactly one indivisible **object**, denoted o_i. The set of objects is $O = \{o_1, o_2, \ldots, o_n\}$.

Each agent i has complete, antisymmetric and transitive preferences R_i over O; that is, R_i is a linear order on O. We write $o_k \, P_i \, o_l$ when $o_k \, R_i \, o_l$ and $o_k \neq o_l$. Preferences are displayed as rankings, so $R_i : o_k, o_l, o_m$ abbreviates $o_k \, P_i \, o_l \, P_i \, o_m$. Let $\mathcal{R}$ denote the set of all linear orders on O. A **preference profile** is a list $R = (R_1, \ldots, R_n) \in \mathcal{R}^N$, and R_{-i} denotes $(R_j)_{j \neq i}$. Since N and O are fixed throughout, a **housing market** is identified with a preference profile $R \in \mathcal{R}^N$.

An **allocation** is a bijection $\mu : N \to O$; μ_i is agent i's allotment. Let $\mathcal{M}$ be the set of allocations. A **mechanism** is a map $\varphi : \mathcal{R}^N \to \mathcal{M}$, and $\varphi_i(R)$ denotes agent i's allotment at $\varphi(R)$.

The TTC allocation of a market R is computed by the TTC algorithm. In each round, every remaining agent points to her most preferred remaining object and every remaining object points to its owner. The resulting digraph has out-degree one at every node and finitely many nodes, so it contains at least one cycle. Every agent on a cycle is assigned the object she points to, and all agents and objects on cycles are removed. The algorithm terminates after at most n rounds. The **TTC mechanism** τ assigns to each market R its TTC allocation $\tau(R)$.

3.2. Axioms

We use the standard axioms. Let φ be a mechanism.

Individual-rationality: for each $R \in \mathcal{R}^N$ and each $i \in N$, $\varphi_i(R) \, R_i \, o_i$.

Pareto-efficiency: for each $R \in \mathcal{R}^N$ there is no allocation $\bar{\mu}$ such that $\bar{\mu}_i \, R_i \, \varphi_i(R)$ for each $i \in N$, with strict preference for some $j \in N$.

Pair-efficiency: for each $R \in \mathcal{R}^N$ there are no distinct $i, j \in N$ such that

$$\varphi_j(R) \, P_i \, \varphi_i(R) \qquad \text{and} \qquad \varphi_i(R) \, P_j \, \varphi_j(R).$$

Pair-efficiency rules out gainful trades between *pairs* of agents only, whereas Pareto-efficiency rules out gainful trades among the members of any subset; so Pareto-efficiency implies pair-efficiency, and the converse fails as soon as $n \geq 3$. This distinction drives the counterexamples of Theorem 2: a pair-efficient mechanism may leave an improvement cycle of length three or more unexecuted.

The next axiom concerns how a mechanism responds when an endowed object becomes more attractive to other agents. Following Biró et al. (2024), we say that a market $\tilde{R}$ is an **improvement for i with respect to** R if the only difference between R and $\tilde{R}$ is that object o_i is ranked weakly higher by the other agents at $\tilde{R}$ than at

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R. Formally: (i) $\tilde{R}_i = R_i$; (ii) for all $j, k \neq i$, if $o_i \, P_j \, o_k$ then $o_i \, \tilde{P}_j \, o_k$; and (iii) for all $j, k, l \neq i$, if $o_k \, R_j \, o_l$ then $o_k \, \tilde{R}_j \, o_l$.

Respecting-improvement: for each $i \in N$ and each pair $R, \tilde{R} \in \mathcal{R}^N$ such that $\tilde{R}$ is an improvement for i with respect to R, $\varphi_i(\tilde{R}) \, R_i \, \varphi_i(R)$.

Respecting-improvement also has an exact adjacent form. Call an improvement adjacent if exactly one agent $j \neq i$ moves o_i upward across the object immediately above it and all other reports remain fixed. Requiring the respecting-improvement inequality only for such adjacent improvements is equivalent to the definition above. Indeed, any improvement can be decomposed into a finite sequence of adjacent upward moves of o_i, possibly in different agents' reports. The owner's preference R_i remains fixed throughout, so transitivity chains the adjacent inequalities. The converse is immediate.

We first record the full incentive requirement before introducing the restriction studied in this paper.

Strategy-proofness: for each $R \in \mathcal{R}^N$, each $i \in N$ and each $R'_i \in \mathcal{R}$,

$$\varphi_i(R) \, R_i \, \varphi_i(R'_i, R_{-i}).$$

We now restrict the class of misreports. The restriction fixes everything about a report except the position of the reporter's own endowment.

Definition 1. Let $i \in N$ and $R_i, R'_i \in \mathcal{R}$. Then R'_i is an **endowment repositioning** of R_i if R_i and R'_i induce the same ranking of the objects other than o_i; that is, for all $o_k, o_l \in O \setminus \{o_i\}$,

$$o_k \, R_i \, o_l \iff o_k \, R'_i \, o_l.$$

Endowment repositioning is an equivalence relation on $\mathcal{R}$. Each of its classes contains exactly n preferences, obtained from a fixed ranking $h_1, h_2, \ldots, h_{n-1}$ of $O \setminus \{o_i\}$ by inserting o_i in one of the n available positions. We call such a class an **endowment-position fiber** for i and index its members by

$$r_i(R_i) \;=\; \big|\{o \in O : o \, P_i \, o_i\}\big| \;\in\; \{0, 1, \ldots, n-1\},$$

the number of objects the agent declares to be better than her own. Thus $r_i = 0$ means the agent reports her endowment as her top object, and $r_i = n - 1$ means she reports it as her worst.

Endowment-position strategy-proofness: for each $R \in \mathcal{R}^N$, each $i \in N$ and each endowment repositioning R'_i of R_i,

$$\varphi_i(R) \, R_i \, \varphi_i(R'_i, R_{-i}).$$

This is the paper's main incentive requirement. It restricts manipulation only within an endowment-position fiber and imposes no condition on reports that rerank non-endowment objects.

Example 1. With three agents, let χ select no trade when agent 1 reports $o_2 \, P_1 \, o_3$, and swap o_1 and o_2 otherwise. The rule is constant on every endowment-position fiber and hence satisfies *endowment-position strategy-proofness*. Yet at $o_2 \, P_1 \, o_1 \, P_1 \, o_3$, the cross-fiber report $o_3 \, P'_1 \, o_2 \, P'_1 \, o_1$ changes agent 1's allotment from o_1 to o_2. Thus χ is manipulable across fibers.

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For the local test of this property, we use the following terminology.

Definition 2. An endowment repositioning R_i' of R_i is **adjacent** if

$$\left|r_i(R_i') - r_i(R_i)\right| = 1.$$

Equivalently, the two reports differ only by transposing o_i and the object immediately above or below it.

Adjacent endowment-position strategy-proofness: for each $R \in \mathcal{R}^N$, each $i \in N$ and each adjacent endowment repositioning R_i' of R_i,

$$\varphi_i(R)\, R_i\, \varphi_i(R_i', R_{-i}).$$

Definition 3. R_i' is a **truncation** of R_i if R_i' is an endowment repositioning of R_i with $r_i(R_i') \leq r_i(R_i)$; equivalently, if $\{o \in O : o\, P_i'\, o_i\} \subseteq \{o \in O : o\, P_i\, o_i\}$.

Endowment-truncation-proofness: for each $R \in \mathcal{R}^N$, each $i \in N$ and each truncation R_i' of R_i, $\varphi_i(R)\, R_i\, \varphi_i(R_i', R_{-i})$. This is the one-sided condition called truncation-proofness by Coreno and Feng (2025, 2026); the qualifier distinguishes it from the assignment-anchored truncation-invariance below.

Truncation-invariance: for each $R \in \mathcal{R}^N$ and each $i \in N$ with $\varphi_i(R) \neq o_i$, if R_i' is the endowment repositioning of R_i that places o_i immediately below $\varphi_i(R)$, then $\varphi_i(R_i', R_{-i}) = \varphi_i(R)$.

Truncation-invariance asks that an agent who declares exactly the objects she is actually assigned or better to be acceptable keeps what she had. It constrains one report per profile, where *endowment-truncation-proofness* constrains the upward half of the fiber and *endowment-position strategy-proofness* constrains all of it.

Upper invariance: for each $R \in \mathcal{R}^N$ and each $i \in N$, put $z = \varphi_i(R)$ and $U_i(z; R_i) = \{o \in O : o\, R_i\, z\}$. If R_i' has the same ordered upper contour through z, in the sense that

$$U_i(z; R_i') = U_i(z; R_i) \quad \text{and} \quad R_i'|_{U_i(z;R_i)} = R_i|_{U_i(z;R_i)},$$

then $\varphi_i(R_i', R_{-i}) = z$. Equivalently, changing only the order of objects below the current allotment leaves that allotment fixed.

Remark 1. Bidirectionality is built into *endowment-position strategy-proofness*. Because "is an endowment repositioning of" is a symmetric relation and the axiom quantifies over all profiles, *endowment-position strategy-proofness* automatically constrains both directions of a report change. The same is true of its adjacent version. For example, applying the adjacent condition at (R_i', R_{-i}) with adjacent repositioning R_i delivers $\varphi_i(R_i', R_{-i})\, R_i'\, \varphi_i(R)$ as well. By contrast, "is a truncation of" is antisymmetric, so *endowment-truncation-proofness* genuinely restricts only upward movements of the endowment. Full-fiber *endowment-position strategy-proofness* may therefore equivalently be read as *bidirectional endowment-truncation-proofness*.

4. Results

TTC is Pareto-efficient and hence pair-efficient; it respects improvement by Biró et al. (2024, Theorem 1); and it is strategy-proof by Roth (1982), hence endowment-position

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strategy-proof. Our main result is the converse.

Theorem 1. *For every $n \geq 3$, the TTC mechanism is the unique mechanism that is pair-efficient, respects improvement, and is endowment-position strategy-proof.*

Relative to Klaus et al. (2025, Theorem 1), Theorem 1 replaces *strategy-proofness* by *endowment-position strategy-proofness*. As the next lemma shows, this condition can be tested through at most two adjacent comparisons at each profile. Thus one distinguished family of report changes identifies TTC and its full and group incentive guarantees.

Lemma 1 (Local-to-global fiber representation)**.** *Fix a mechanism φ, an agent i, a profile R_{-i}, and a ranking $h_1, h_2, \ldots, h_{n-1}$ of $O \setminus \{o_i\}$; let R_i^r be the member of the corresponding fiber with $r_i = r$ and write $x(r) = \varphi_i(R_i^r, R_{-i})$. The following are equivalent on this fiber:*

(a) φ satisfies all endowment-position constraints;

(b) φ satisfies the adjacent endowment-position constraints;

(c) one of the following forms holds:

(i) $x(r) = z$ for every r, for some $z \in O$; or

(ii) there is a unique $k \in \{1, \ldots, n-1\}$ with $x(r) = o_i$ for $r < k$ and $x(r) = h_k$ for $r \geq k$.

If φ is individually-rational, the constant form in (c)(i) is possible only for $z = o_i$.

Proof. Assume the adjacent constraints in part (b). For $r \in \{0, \ldots, n-2\}$, put $a = x(r)$ and $b = x(r+1)$. Applying the adjacent axiom with each of R_i^r and R_i^{r+1} as the true report gives

$$a \, R_i^r \, b \qquad \text{and} \qquad b \, R_i^{r+1} \, a.$$

If $a \neq b$, both comparisons are strict. The two reports differ only in the relative order of o_i and h_{r+1}, so that is the only pair of objects whose order is reversed. It follows that

$$a = o_i, \qquad b = h_{r+1}.$$

Thus every neighboring pair of assignments is either constant or makes the single forward switch just displayed.

If no edge makes that switch, x is constant. If the edge from $k-1$ to k makes it, then $x(k-1) = o_i$ and $x(k) = h_k$. Every earlier edge must be constant, working backwards, because a nonconstant edge would end at a non-endowment object; every later edge must be constant, working forwards, because a nonconstant edge would start at o_i. Hence $x(r) = o_i$ for $r < k$ and $x(r) = h_k$ for $r \geq k$. This proves (b)⇒(c).

A constant map plainly satisfies all endowment-position constraints. A threshold map does as well: below the threshold the agent receives o_i and ranks it above h_k, while at and above the threshold she receives h_k and ranks it above o_i; reports on the same side give the same assignment. Thus (c)⇒(a). Finally, (a)⇒(b) is immediate because adjacent repositionings are a subset of all endowment repositionings.

If φ is individually-rational, then at R_i^0 the endowment is the top object, so $x(0) = o_i$. This excludes every constant map except the one at o_i; all of the remaining threshold maps are individually-rational. □

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Local constraints therefore propagate along the entire fiber without a participation assumption. There are exactly $2n-1$ allowable fiber maps: the n constant maps and the $n-1$ threshold maps. Individual-rationality excludes a constant outside object, so the agent then faces a two-element menu — her endowment and one fixed outside object h_k — and receives the outside object exactly when she declares it acceptable. As far as *individual-rationality* and *endowment-position strategy-proofness* are concerned, thresholds remain unlinked across fibers. Appendix A coordinates this freedom for every $n \geq 4$; Theorem 2 shows that the three-agent case remains unique.

The local test also clarifies the logical relations among the incentive requirements used in the paper.

Proposition 1. *Endowment-position strategy-proofness is equivalent to its apparently weaker adjacent version. Strategy-proofness implies endowment-position strategy-proofness, which implies endowment-truncation-proofness and truncation-invariance. None of these three implications can be reversed; the failure of the reverse implication from truncation-invariance holds even among individually-rational mechanisms. Among individually-rational mechanisms, upper invariance implies endowment-position strategy-proofness, and this implication is strict.*

The equivalence is immediate from Lemma 1. The strict reverse implications use the two mechanisms introduced next, so their verification is completed after those examples.

The restriction is sharp in two natural directions. First, Klaus et al. (2025) establish that *strategy-proofness* cannot be dropped from their theorem by means of the following mechanism. Let $n = 3$ and let $\bar{R}$ be the profile

$$\bar{R}_1 : o_2, o_1, o_3 \qquad \bar{R}_2 : o_3, o_2, o_1 \qquad \bar{R}_3 : o_1, o_3, o_2,$$

and define $\beta(\bar{R}) \equiv (o_1, o_2, o_3)$, the endowment allocation, and $\beta(R) \equiv \tau(R)$ for every $R \neq \bar{R}$. They show that β is pair-efficient and respects improvement but is not strategy-proof.

Proposition 2. *The mechanism β is individually-rational, pair-efficient, respects improvement, and is endowment-truncation-proof, and $\beta \neq \tau$. Consequently, pair-efficiency, respecting-improvement and endowment-truncation-proofness do not characterize the TTC mechanism, and the failure occurs already for $n = 3$.*

Proof. Away from $\bar{R}$ the mechanism coincides with TTC. At $\bar{R}$ every agent ranks exactly one object above her endowment, and these three desired objects form the directed cycle o_2, o_3, o_1. Thus no two agents prefer each other's endowments, so the endowment allocation $\beta(\bar{R})$ is pair-efficient. It is also individually-rational. Since TTC has both properties, so does β at every profile.

We next verify respecting-improvement. Fix an owner i and suppose that $\widetilde{R}$ is an improvement for i with respect to R. If neither endpoint is $\bar{R}$, the conclusion follows from respecting-improvement of TTC. Suppose first that $R = \bar{R}$ and $\widetilde{R} \neq \bar{R}$. Then $\beta_i(R) = o_i$, while $\beta(\widetilde{R}) = \tau(\widetilde{R})$. Because $\widetilde{R}_i = \bar{R}_i$, individual rationality of TTC gives

$$\beta_i(\widetilde{R}) \; \bar{R}_i \; o_i = \beta_i(\bar{R}).$$

Suppose instead that $\widetilde{R} = \bar{R}$ and $R \neq \bar{R}$. Of the two agents other than i, exactly one, say p, ranks o_i first at $\bar{R}$; she ranks her own endowment o_p second. The other

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ranks o_i last. Hence the only way for $\bar{R}$ to be an improvement for i relative to a distinct R is that o_i was lower in R_p. The relative order of the other objects is fixed, so p ranks o_p first at R and exits in a singleton TTC cycle. Of the two remaining agents, the one different from i then also ranks her own remaining endowment first and exits, leaving i with o_i. Therefore

$$\beta_i(R) = \tau_i(R) = o_i = \beta_i(\bar{R}),$$

which completes the respecting-improvement argument.

It remains to prove endowment-truncation-proofness. A fiber not meeting $\bar{R}$ is governed throughout by TTC. On a fiber meeting $\bar{R}$, let h_i denote the object ranked above o_i there and let ℓ_i be the remaining object. The three possible reports and assignments are

r_i	$R_i^{r_i}$	$\beta_i(R_i^{r_i}, \bar{R}_{-i})$
0	o_i, h_i, ℓ_i	o_i
1	$\bar{R}_i:\ h_i, o_i, \ell_i$	o_i
2	h_i, ℓ_i, o_i	h_i.

The first row is a singleton TTC cycle. In the last row the original three-cycle is restored, so TTC assigns h_i. A truncation from row 1 to row 0 leaves the assignment unchanged; a truncation from row 2 to either lower row replaces the agent's top object h_i by her endowment, which cannot be profitable at R_i^2. Thus β is endowment-truncation-proof.

Finally, TTC executes the three-cycle at $\bar{R}$, so $\tau(\bar{R}) = (o_2, o_3, o_1) \neq \beta(\bar{R})$. This proves every assertion. □

The mechanism β makes the missing downward discipline transparent. At $\bar{R}$ every agent ranks her own endowment second, so no agent can *truncate* into a profitable deviation: raising o_i to the top only makes the agent's own object her declared first choice. What each agent can do is move her endowment *down*, which leaves the exceptional profile $\bar{R}$ and returns her to TTC and to her top object. This deviation is invisible to *endowment-truncation-proofness* and is forbidden by *endowment-position strategy-proofness*.

This also locates the role of the efficiency requirement. At $\bar{R}$, the three-cycle (o_2, o_3, o_1) is a Pareto improvement over no trade, so β is not Pareto-efficient. A one-sided truncation condition can characterize TTC when paired with Pareto-efficiency (Coreno and Feng, 2026); after efficiency is weakened to pair-efficiency, that three-cycle is no longer detected, and the downward direction of the incentive condition closes the gap.

Truncation-invariance supplies a different local weakening: under *individual-rationality* it tests only the report that places the endowment immediately below the assignment. Proposition 3 gives the corresponding boundary.

Proposition 3. *At $n = 4$, Pareto-efficiency, respecting-improvement and truncation-invariance do not characterize the TTC mechanism.*

Proof. Let $n = 4$ and let $R^\dagger$ be the profile

$$R_1^\dagger : o_1, o_2, o_3, o_4, \qquad R_2^\dagger : o_1, o_3, o_2, o_4,$$
$$R_3^\dagger : o_1, o_4, o_2, o_3, \qquad R_4^\dagger : o_1, o_3, o_2, o_4.$$

At $R^\dagger$, TTC first removes agent 1 and then executes the cycle $(3, 4)$, so, writing

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$y := \tau(R^\dagger)$, define

$$y = (o_1, o_2, o_4, o_3), \qquad x := \gamma(R^\dagger) = (o_1, o_3, o_4, o_2),$$
$$\gamma(R) = \tau(R) \quad (R \neq R^\dagger).$$

Thus $\gamma \neq \tau$. Since TTC is individually-rational and Pareto-efficient, only $R^\dagger$ must be checked for those two properties. Individual-rationality follows directly from the four displayed rankings. To see Pareto-efficiency, any allocation weakly improving every agent over x must leave agent 1 with o_1, her first choice. With o_1 unavailable, agent 2 must keep o_3, her first choice among the remaining objects. With o_1 and o_3 unavailable, agent 3 must keep o_4, which she prefers to either remaining object. Feasibility then leaves o_2 to agent 4. Thus no distinct allocation Pareto dominates x.

We next prove respecting-improvement. Fix i and suppose that $\widetilde{R}$ is an improvement for i with respect to R. If neither endpoint is $R^\dagger$, the claim follows from respecting-improvement of TTC. At the exceptional profile,

$$\begin{aligned} &x_1 = y_1 = o_1, && x_3 = y_3 = o_4, \\ &x_2 = o_3 \; P_2^\dagger \; y_2 = o_2, && y_4 = o_3 \; P_4^\dagger \; x_4 = o_2. \end{aligned} \tag{1}$$

For an improvement leaving $R^\dagger$, TTC compares the new assignment with y_i; for one entering $R^\dagger$, it compares y_i with the old assignment. Hence (1) and TTC's axiom settle every incident pair except

$$\big(R = R^\dagger,\ i = 2\big) \qquad \text{and} \qquad \big(\widetilde{R} = R^\dagger,\ i = 4\big).$$

In the first case, only o_2 may move upward in the reports of agents $1, 3, 4$. If it passes o_1 in agent 1's report, TTC executes $(1, 2)$ and gives agent 2 object o_1. Otherwise agent 1 first removes o_1. If agent 3's report changes, the residual cycle is $(2, 3)$; if not, agent 4's report must change and the residual cycle is $(2, 3, 4)$. Either way agent 2 receives o_3. Thus $\gamma_2(\widetilde{R})\, R_2^\dagger\, \gamma_2(R^\dagger)$.

In the second case, $R^\dagger$ is an improvement for agent 4 relative to $R \neq R^\dagger$. Object o_4 can have been lower only in agent 3's report, where it lies below o_2. After agent 1 removes o_1, TTC executes $(2, 3)$ and agent 4 keeps o_4. Therefore

$$\gamma_4(R^\dagger) = o_2 \; P_4^\dagger \; o_4 = \gamma_4(R),$$

which completes the respecting-improvement proof.

Finally, consider truncation-invariance. First note that TTC has this property. If $z = \tau_i(R) \neq o_i$, let R_i' place o_i immediately below z, and put $z' = \tau_i(R_i', R_{-i})$. Strategy-proofness at the two reports gives $z\, R_i\, z'$ and $z'\, R_i'\, z$. Since the reports agree on all non-endowment objects and $z\, P_i'\, o_i$, these inequalities force $z' = z$.

It remains to consider comparisons incident to $R^\dagger$. Starting there, the condition is void for agent 1, while the prescribed report is unchanged for agents 2 and 4. For agent 3 it is R_3' : o_1, o_4, o_3, o_2. After agent 1 exits, TTC still executes $(3, 4)$, so $\gamma_3(R_3', R_{-3}^\dagger) = o_4 = x_3$.

No distinct source profile can have $R^\dagger$ as its prescribed target. Indeed, fix $R_{-i}^\dagger$, and let R_i^r be the report in the fiber of $R_i^\dagger$ with $r = 0, 1, 2, 3$ objects above o_i. Direct TTC

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analysis gives, for $i = 1, 2, 3, 4$ respectively,

$$(o_1, o_2, o_2, o_2), \quad (o_2, o_2, o_2, o_2), \quad (o_3, o_3, o_4, o_4), \quad (o_4, o_4, o_3, o_3). \tag{2}$$

These are TTC assignments. The entry at $R^\dagger$ itself plays no role here, because the source profile in the remaining comparison is distinct from $R^\dagger$. For $R_i^\dagger$ to be the prescribed target, the source assignment would have to be the object immediately above o_i there. No such object exists for $i = 1$; for $i = 2, 3, 4$, these objects are o_3, o_2, o_2, respectively, and none occurs in the corresponding tuple in (2). Thus every remaining comparison has both endpoints off $R^\dagger$ and inherits truncation-invariance from TTC. □

The two counterexamples also supply the strict reverse implications in Proposition 1.

Proof of Proposition 1. Every endowment repositioning is an admissible report under strategy-proofness, and every truncation is an endowment repositioning. Hence strategy-proofness implies *endowment-position strategy-proofness*, which implies *endowment-truncation-proofness*. The equivalence of *adjacent endowment-position strategy-proofness* and *endowment-position strategy-proofness* is Lemma 1, applied to every agent, every profile of the other agents and every endowment-position fiber; it requires no individual-rationality assumption.

The implication from strategy-proofness to the equivalent endowment-position conditions is strict. The mechanism χ in Example 1 is constant on every reporter's endowment-position fiber, yet the displayed cross-fiber report is profitable. Thus it satisfies *endowment-position strategy-proofness* but is not strategy-proof.

The reverse implication from *endowment-truncation-proofness* is false by Proposition 2. Indeed, β satisfies *endowment-truncation-proofness*, but at $\bar{R}$ agent 1 receives o_1 and can report o_2, o_3, o_1, an endowment repositioning, to leave the exceptional profile and receive o_2 from TTC. The report moves o_1 from second place to last. Thus β violates *endowment-position strategy-proofness*; the violation happens to be adjacent.

To obtain truncation-invariance, let $z = \varphi_i(R) \neq o_i$, let R_i' place o_i immediately below z, and write $w = \varphi_i(R_i', R_{-i})$. Applying *endowment-position strategy-proofness* at the two profiles gives

$$z \, R_i \, w \qquad \text{and} \qquad w \, R_i' \, z.$$

Since $z \, P_i' \, o_i$, the second inequality rules out $w = o_i$. Both z and w are therefore non-endowment objects, whose relative order is the same in R_i and R_i'. The two inequalities then force $w = z$. Hence *endowment-position strategy-proofness* implies truncation-invariance.

The reverse implication from truncation-invariance fails even with individual rationality. The mechanism γ of Proposition 3 has both properties, but it is not endowment-position strategy-proof. Hold $R_{-2}^\dagger$ fixed and let agent 2 report $R_2^3 : o_1, o_3, o_4, o_2$. The corresponding TTC fiber row displayed in the proof of Proposition 3 gives her o_2. By repositioning only her endowment to report $R_2^\dagger : o_1, o_3, o_2, o_4$, she reaches $R^\dagger$ and receives o_3 under γ; yet $o_3 \, P_2^3 \, o_2$. The move changes r_2 from 3 to 2, so it is a profitable adjacent endowment repositioning and proves strictness.

We next compare upper invariance with *endowment-position strategy-proofness*. Fix a fiber as in Lemma 1. Suppose $x(r) = h_j$ for some r. Individual rationality gives $j \leq r$. For every $r' \geq j$, the reports R_i^r and $R_i^{r'}$ have the same ordered upper contour through

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h_j, so upper invariance gives $x(r') = h_j$. If $r' < j$ and $x(r') = h_a \neq o_i$, individual rationality gives $a \leq r' < j$. The reports $R_i^{r'}$ and R_i^j then have the same ordered upper contour through h_a, so upper invariance would give $x(j) = h_a$, contradicting $x(j) = h_j$. Hence $x(r') = o_i$ for every $r' < j$. If the fiber has no non-endowment assignment, individual rationality makes it constant at o_i. Every fiber therefore has one of the forms in Lemma 1(c), which proves *endowment-position strategy-proofness*.

The implication is strict already with four agents. Define δ to swap o_1 and o_2 exactly when $o_2 \, P_1 \, o_1$, $o_1 \, P_2 \, o_2$, and $o_3 \, P_1 \, o_4$, and otherwise to select the endowment allocation. The mechanism is individually-rational. On every fiber it is either constant at the endowment or has the threshold form of Lemma 1, so it satisfies *endowment-position strategy-proofness*. Fix a profile at which agent 2 ranks o_1 above o_2 and agent 1 reports o_2, o_1, o_4, o_3. Agent 1 receives o_1. Merely swapping o_3 and o_4, both below that allotment, changes her report to o_2, o_1, o_3, o_4 and makes δ assign her o_2. Thus δ violates upper invariance. □

Remark 2. Truncation-invariance pins the assignment only at the single report that places o_i immediately below it. Endowment-position strategy-proofness yields the following stronger invariance. If $z = \varphi_i(R) \neq o_i$, then at every endowment repositioning R_i' of R_i that still ranks z above o_i,

$$\varphi_i(R_i', R_{-i}) = z. \tag{*}$$

Indeed, writing the new allotment as w, *endowment-position strategy-proofness* at the two endpoints gives $z \, R_i \, w$ and $w \, R_i' \, z$. The latter comparison excludes $w = o_i$ because $z \, P_i' \, o_i$; and the two reports agree on the order of all non-endowment objects, so the two comparisons force $w = z$.

Conversely, under *individual-rationality*, (*) yields exactly the forms in Lemma 1(c). It permits at most one non-endowment object: if h_a and h_b, with $a < b$, were both assigned, then *individual-rationality* at the report assigning h_b places both above o_i, so (*) from a report assigning h_a would require h_a there, a contradiction. If h_a occurs, *individual-rationality* assigns o_i below position a and (*) assigns h_a from position a onward; if none occurs, *individual-rationality* makes the map constant at o_i. Lemma 1 therefore gives *endowment-position strategy-proofness*. Thus (*) is the full-fiber version of the one-target invariance required by truncation-invariance.

Because *pair-efficiency* and *respecting-improvement* imply *individual-rationality* (Klaus et al., 2025, Lemma 1), the next result isolates what *respecting-improvement* adds beyond participation and locates the exact population boundary.

Theorem 2. *For $n = 3$, the TTC mechanism is the unique mechanism satisfying individual-rationality, pair-efficiency and endowment-position strategy-proofness. For every $n \geq 4$ this is false: there is a mechanism satisfying the three axioms and differing from TTC.*

Section 5 proves the positive part; Appendix A constructs the counterexample family for every $n \geq 4$. The contrast identifies the added role of *respecting-improvement*: it changes the report of an agent *other* than the owner and thereby links the endowment-position fibers on which *endowment-position strategy-proofness* acts.

Finally, the characterization is tight in the standard logical sense.

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Proposition 4. *The three axioms in Theorem 1 are logically independent.*

(a) The no-trade mechanism respects improvement and is strategy-proof, hence endowment-position strategy-proof, but is not pair-efficient.

(b) There is a serial dictatorship that is Pareto-efficient, hence pair-efficient, and strategy-proof, hence endowment-position strategy-proof, but does not respect improvement.

(c) The mechanism β of Proposition 2 is pair-efficient and respects improvement but is not endowment-position strategy-proof.

Proof. (a) The no-trade mechanism assigns o_i to i at every profile. It is constant, hence strategy-proof and so, by Proposition 1, endowment-position strategy-proof; and no agent's allotment ever changes, so it respects improvement. It is not pair-efficient: at any profile at which two agents rank each other's endowments first, both would gain by exchanging.

(b) Fix any $n \geq 3$ and let ζ be the serial dictatorship with order $1, 2, \ldots, n$. Consider a profile with

$$R_1 : o_1, o_2, o_3, \ldots, o_n, \qquad R_2 : o_2, o_1, o_3, \ldots, o_n,$$

and let every agent $k \geq 3$ rank o_k first. At this profile agents 1 and 2 receive o_1 and o_2, respectively. Now replace only agent 1's report by

$$\tilde{R}_1 : o_2, o_1, o_3, \ldots, o_n,$$

leaving every other report unchanged. The new profile is an improvement for agent 2: only o_2 has moved upward in another agent's ranking, and the relative order of all other objects is unchanged. Under the new profile the first dictator takes o_2, and agent 2 receives o_1, which she ranks below o_2. Thus ζ does not respect improvement. Serial dictatorship is Pareto-efficient and strategy-proof, and therefore is pair-efficient and endowment-position strategy-proof. This embeds the two-agent witness of Klaus et al. (2025, Example 2) in every population $n \geq 3$.

(c) The mechanism β of Proposition 2 is pair-efficient and respects improvement. It is not endowment-position strategy-proof: at $\bar{R}$ agent 1 receives o_1, while reporting $\bar{R}'_1 : o_2, o_3, o_1$ — which moves only her own endowment, from second place to last — leaves the exceptional profile, so the mechanism returns τ and gives her o_2, and $o_2 \, \bar{P}_1 \, o_1$. By symmetry the same deviation is available to agents 2 and 3. □

5. Proof of the main theorem

The proof has five stages. We choose a minimal conflict profile, derive the first failed cycle's canonical structure and show that the cycle is proper. A common-sink construction excludes past objects, downward pinning separates the cycle, and Lemma 5 then induces the smaller counterexample that strong induction excludes.

Fix a selector that resolves ties between simultaneous TTC cycles — say, always execute first the cycle containing the lowest-indexed agent. Running the cycles one at a time in that order gives the same allocation as the simultaneous description in Section 3. This indexes the proof without changing TTC's allocation; every reference

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below to an earlier cycle or conflict uses this fixed ordering. Write

$$C_1(R),\, C_2(R),\, \ldots,\, C_{q(R)}(R)$$

for the resulting sequence of cycles, and let $O_t(R)$ be the set of objects still present when $C_t(R)$ is formed. For $i \in C_t(R)$ let $c_i(R)$ denote the object i points to in $C_t(R)$, that is, i's most preferred object in $O_t(R)$. Say that φ **executes** $C_t(R)$ if $\varphi_i(R) = c_i(R)$ for every $i \in C_t(R)$, and put

$$\rho(R) = \min\{t : \varphi \text{ does not execute } C_t(R)\},$$
$$s(R) = \sum_{i \in N} \big|\{o \in O : o\, P_i\, o_i\}\big|,$$

with $\rho(R) = \infty$ when $\varphi(R) = \tau(R)$. We call R a **conflict profile** if $\rho(R) < \infty$. Here ρ records the earliest departure from TTC, while s counts all objects agents report above their own endowments. Every upward endowment repositioning other than the report itself strictly lowers s, providing the second tie-break in the minimization below.

We begin with the observation of Klaus et al. (2025, Lemma 1) that pair-efficiency and respecting-improvement imply individual rationality. Fix Q and i; we show $\varphi_i(Q)\, Q_i\, o_i$. Let R be obtained from Q by making o_i the worst object for every agent $j \neq i$, leaving the relative ranking of all other objects and all of Q_i unchanged; thus $R_i = Q_i$ and Q is an improvement for i with respect to R. Respecting-improvement gives $\varphi_i(Q)\, R_i\, \varphi_i(R)$, so it suffices to show $\varphi_i(R)\, R_i\, o_i$.

If $\varphi_i(R) = o_i$ there is nothing to prove. Otherwise $\varphi_i(R) = o_j$ for some $j \neq i$; let k be the agent with $\varphi_k(R) = o_i$, so $k \neq i$. By construction o_i is last in R_k, whence $o_j\, P_k\, o_i = \varphi_k(R)$. Were $o_i\, P_i\, o_j = \varphi_i(R)$, agents i and k would both strictly gain by exchanging their assignments, contradicting pair-efficiency. Hence $\varphi_i(R) = o_j\, R_i\, o_i$, as required. Thus φ is individually-rational. We next use respecting-improvement to connect endowment-position fibers.

5.1. Minimal conflict

For the rest of this section suppose, for contradiction, that $\varphi \neq \tau$ satisfies the three axioms of Theorem 1; by the preceding argument it is also individually-rational. We use *endowment-position strategy-proofness* throughout the remainder of the proof. Choose a conflict profile R^* that first minimizes $\rho(R)$ and then, among those, minimizes $s(R)$. Write $t = \rho(R^*)$ and $C = C_t(R^*) = (i_0, i_1, \ldots, i_{m-1})$. For $i \in C$ abbreviate $c_i := c_i(R^*)$, and index C so that $c_{i_k} = o_{i_{k+1}}$ with indices modulo m.

We use repeatedly the following observation. If R_i' is obtained from R_i^* by moving o_i up to a position still below c_i, then the first t TTC cycles are unchanged: every object i ranks above c_i has already been removed before stage t, so i's pointer in each of the first t rounds is unaffected, and no other agent's report has changed.

Lemma 2 (Canonical first-conflict structure)**.** *For every $i \in C$, $\varphi_i(R^*) = o_i$, and c_i is ranked immediately above o_i in R_i^*.*

Proof. Since φ does not execute C, some $i \in C$ has $\varphi_i(R^*) \neq c_i$. As φ executes $C_1, \ldots, C_{t-1}$, agent i receives an object of $O_t(R^*)$, and c_i is her most preferred such object, so $c_i\, P_i^*\, \varphi_i(R^*)$. Individual rationality gives $\varphi_i(R^*)\, R_i^*\, o_i$.

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Suppose $\varphi_i(R^*) \neq o_i$, so that $c_i \, P_i^* \, \varphi_i(R^*) \, P_i^* \, o_i$. Let R_i' move o_i up to the position immediately below c_i and set $R' = (R_i', R_{-i}^*)$. This is an endowment repositioning (indeed, a truncation), so *endowment-position strategy-proofness* applies, and $s(R') < s(R^*)$.

If $\varphi_i(R') = c_i$, then at the true preference R_i^* agent i gains by reporting R_i', since $c_i \, P_i^* \, \varphi_i(R^*)$; this contradicts *endowment-position strategy-proofness*. Hence $\varphi_i(R') \neq c_i$, so φ does not execute $C_t(R') = C$ and R' is a conflict profile with $\rho(R') \leq t$. Minimality of ρ gives $\rho(R') \geq t$, hence $\rho(R') = t$, and then $s(R') < s(R^*)$ contradicts the minimality of s. Therefore $\varphi_i(R^*) = o_i$.

Now let p be the predecessor of i on C, so $c_p = o_i$. Since i holds o_i, agent p does not receive c_p, and the argument just given applies verbatim to p. Propagating around C establishes the first conclusion.

For the second, fix $i \in C$. We have $\varphi_i(R^*) = o_i$ and $c_i \, P_i^* \, o_i$. Suppose some object lies strictly between them, and let R_i' move o_i up to the position immediately below c_i, with $R' = (R_i', R_{-i}^*)$; then $s(R') < s(R^*)$.

Applying *endowment-position strategy-proofness* at R^* gives $o_i = \varphi_i(R^*) \, R_i^* \, \varphi_i(R')$. Applying individual rationality at R' gives $\varphi_i(R') \, R_i' \, o_i$. Every object other than o_i that is weakly preferred to o_i under R_i' is strictly preferred to o_i under R_i^*, since R_i' only moved o_i upwards. The two displayed requirements are therefore compatible only if $\varphi_i(R') = o_i$.

Hence $\varphi_i(R') \neq c_i$, so $\rho(R') \leq t$; minimality of ρ gives $\rho(R') = t$, and $s(R') < s(R^*)$ contradicts the minimality of s. □

Lemma 2 says that at R^* the mechanism returns no trade on C, and that each member of C finds exactly one object of $O_t(R^*)$ acceptable besides her own, namely her TTC successor. We record the consequence we use.

Claim 1. *Let R agree with R^* except possibly in the strict lower contour of o_{i_0} in R_{i_0}: the ordered upper contour through o_{i_0} is unchanged, and only objects strictly below o_{i_0} may be permuted. Suppose φ executes $C_r(R)$ for every $r < t$. Then either every $i \in C$ receives o_i at R, or every $i \in C$ receives c_i.*

Proof. First, the TTC cycles through stage t are the same at R and R^*. Indeed, induct over those stages. Until C is reached, c_{i_0} remains available and is ranked above o_{i_0}, so permuting the strict lower contour of o_{i_0} does not change i_0's pointer; every other report is unchanged. The fixed selector therefore removes the same cycle at each stage.

Because φ executes the first $t-1$ cycles, their members occupy all objects outside $O_t(R) = O_t(R^*)$. Feasibility puts every member of C in $O_t(R^*)$. For agents other than i_0, individual rationality, residual optimality and Lemma 2 leave only o_i or c_i. The same menu applies to i_0 because her ordered upper contour through o_{i_0} is unchanged. No agent outside C can receive an object of C: if $h \notin C$ received o_{i_k}, then i_{k-1} could not receive $c_{i_{k-1}}$ and so would take $o_{i_{k-1}}$, while i_k, whose endowment is gone, would take c_{i_k}, forcing $i_{k+1}, \ldots, i_{k-2}$ in turn to take their successors, so that i_{k-2} would claim $o_{i_{k-1}}$ — already held by i_{k-1}. Within C the same propagation shows that if one member takes her successor then all do. □

Lemma 3 (Downward pinning)**.** *Let R be an improvement for j with respect to Q, and suppose $\varphi_j(R) = o_j$. Then $\varphi_j(Q) = o_j$.*

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Proof. Respecting-improvement gives $o_j = \varphi_j(R)\, R_j\, \varphi_j(Q)$, and individual rationality at Q gives $\varphi_j(Q)\, R_j\, o_j$. Since $R_j = Q_j$ and preferences are strict, $\varphi_j(Q) = o_j$. □

Let A_t denote the agents active when $C = C_t(R^*)$ is formed, and write $O_C = \{o_i : i \in C\}$.

Claim 2. *The minimal conflict profile has the following two properties.*

(a) If $h \notin C$ is active at stage t and a_h is her most preferred object in $O_t(R^*)$, there is no $j \in C$ with $a_h\, P_h^*\, o_j\, P_h^*\, o_h$.

(b) If $x \notin A_t$, then $o_x\, P_x^*\, o_j$ for every $j \in C$.

Proof. For part (a), suppose such a j exists. Move o_j down in R_h^* by adjacent transpositions until it sits immediately above o_h, obtaining profiles $R^0 = R^*, R^1, \ldots, R^q$. Every position touched lies strictly below a_h, so h's pointer is unchanged in each of the first t rounds and $C_r(R^\ell) = C_r(R^*)$ for $r \le t$ and every ℓ.

By Lemma 2, $\varphi_j(R^0) = o_j$. Each $R^{\ell-1}$ is an improvement for j with respect to R^ℓ, since the only change is that o_j stands one place higher in R_h; so Lemma 3 gives $\varphi_j(R^\ell) = o_j$ for every ℓ. As $c_j \neq o_j$, the mechanism does not execute C at R^ℓ, so R^ℓ is a conflict profile with $\rho(R^\ell) \le t$, and minimality of ρ gives $\rho(R^\ell) = t$. Note that $s(R^\ell) = s(R^*)$ throughout, because o_j remains above o_h.

Finally transpose o_j and o_h, obtaining $\hat{R}$. This again moves o_j down one place, so Lemma 3 gives $\varphi_j(\hat{R}) = o_j$ and, as before, $\rho(\hat{R}) = t$. But o_h has moved up one place, so $s(\hat{R}) = s(R^*) - 1$, contradicting the minimality of s.

For part (b), suppose instead that $o_j\, P_x^*\, o_x$ for some $j \in C$. Let x be removed in a cycle before stage t, and let ν_x be the object to which she points in that cycle. Since every object of O_C remains available until stage t, $\nu_x\, R_x^*\, o_j$; equality is impossible because assigning o_j in that earlier cycle would remove j before stage t. Thus $\nu_x\, P_x^*\, o_j$. Move o_j in R_x^* to the position immediately below o_x, obtaining Q. At every stage before x is removed, her current top object is weakly preferred to ν_x and hence strictly preferred to o_j. Her pointers before removal are therefore unchanged; after removal her report is irrelevant. Thus the first t TTC cycles are unchanged.

The original profile is an improvement for j with respect to Q. Lemma 2 gives $\varphi_j(R^*) = o_j$, so Lemma 3 gives $\varphi_j(Q) = o_j$. Since $C_t(Q) = C$ and $c_j \neq o_j$, the profile Q is a conflict profile with $\rho(Q) \le t$. Minimality gives $\rho(Q) = t$. But o_j has crossed o_x, so $s(Q) = s(R^*) - 1$, contradicting the second minimization. □

Part (a) says that whatever separates a stage-t agent from her best remaining object, it is not owned by a member of the first failed cycle.

Call C **tail-isolated at** i_0 if, in $R_{i_0}^*$ restricted to $O_t(R^*)$, the objects $o_{i_2}, \ldots, o_{i_{m-1}}$ occupy the $m-2$ positions immediately below o_{i_0}, in some order, and every object of $O_t(R^*) \setminus \{o_{i_0}, \ldots, o_{i_{m-1}}\}$ is ranked below all of them.

By Lemma 2, $o_{i_1} = c_{i_0}$ sits immediately above o_{i_0}, so tail-isolation says exactly that the cycle occupies a contiguous block at the top of i_0's residual ranking. When C exhausts $O_t(R^*)$ the condition holds vacuously.

Lemma 4. *C is a proper subset of the agents still active at stage t.*

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Proof. By Lemma 2, a failed cycle cannot have length one: if $m = 1$, then $c_{i_0} = o_{i_0} = \varphi_{i_0}(R^*)$, so the cycle is executed. Nor can $m = 2$, because Lemma 2 leaves both members with their endowments although each strictly prefers her successor, contradicting pair-efficiency. Hence $m \geq 3$.

If C exhausted the active market, it would be tail-isolated at every member. Suppose, more generally, that it is tail-isolated at i_0; we derive a contradiction.

Step 1: stably sort the residual tail. Because tail-isolation is a statement about the restriction to $O_t(R^*)$, past objects may be interspersed with the tail objects in the full ranking. To obtain the target order, process the tail from bottom to top. First lower $o_{i_{m-1}}$ until it is last among the tail objects in the restriction to $O_t(R^*)$; then, for $r = m-2, \ldots, 2$, lower o_{i_r} until it lies immediately above the already fixed suffix $o_{i_{r+1}}, \ldots, o_{i_{m-1}}$ in that restriction. Skip a move if the object is already in place, and preserve the relative order of every other object at each nontrivial move. Tail-isolation ensures that every such move is downward and crosses neither o_{i_0} nor an active outsider object; it may cross past objects as well as other tail objects. The resulting finite sequence of stable downward moves in R_{i_0} ends at

$$R_{i_0}|_{O_t(R^*)}: \quad o_{i_1} \succ o_{i_0} \succ o_{i_2} \succ \cdots \succ o_{i_{m-1}} \succ O_t(R^*) \setminus O_C.$$

This order places the relevant predecessor's endowment above the reporter's putative assignment, as the pair-efficiency comparison in Step 4 requires. Every moved object remains below o_{i_0} throughout.

We show inductively that every member of C keeps her endowment after each move. Suppose the moved object is o_j. The profile before the move is an improvement for owner j relative to the profile after it, so Lemma 3 pins j at o_j. The edit is confined to R_{i_0} below o_{i_0} and hence leaves the first t TTC cycles and their within-cycle pointers unchanged. Since $c_j \neq o_j$, the profile after the move is a conflict profile with $\rho \leq t$. Minimality of $\rho(R^*) = t$ therefore implies that the first $t-1$ cycles are executed. Claim 1 applies, and the fact that j holds her endowment forces every member of C to hold her endowment. Let $\hat{R}$ be the terminal profile. Then

$$\varphi_i(\hat{R}) = o_i \qquad (i \in C), \tag{3}$$

and

$$\hat{R}_{i_0}|_{O_t(R^*)}: \quad o_{i_1} \succ o_{i_0} \succ o_{i_2} \succ \cdots \succ o_{i_{m-1}} \succ O_t(R^*) \setminus O_C. \tag{4}$$

Step 2: move the reporter's endowment to the foot of the cycle block. Let $\tilde{R}$ be obtained from $\hat{R}$ by moving o_{i_0}, in the full report of i_0, to the position immediately below $o_{i_{m-1}}$, while preserving the relative order of all non-endowment objects. This is an endowment repositioning. The first t TTC cycles remain unchanged: every comparison altered by the move lies below $c_{i_0} = o_{i_1}$.

If $\tilde{R}$ is not a conflict profile, then, by definition, $\varphi(\tilde{R}) = \tau(\tilde{R})$ and $\varphi_{i_0}(\tilde{R}) = o_{i_1}$, so we pass directly to Step 5. Suppose instead that $\tilde{R}$ is a conflict profile. Minimality of $\rho(R^*)$ implies $\rho(\tilde{R}) \geq t$, so φ executes the first $t-1$ TTC cycles at $\tilde{R}$. The cycle assignments are then confined as follows.

The members of the first $t-1$ cycles receive exactly all past objects $O \setminus O_t(R^*)$; feasibility therefore prevents every member of C from receiving a past object. For each $h \in C \setminus \{i_0\}$, the report is unchanged from R_h^*, so individual rationality, residual

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optimality of c_h, and Lemma 2 imply

$$\varphi_h(\tilde{R}) \in \{o_h, c_h\}. \tag{5}$$

By (4), after the endowment move every active outsider object lies below o_{i_0} in $\tilde{R}_{i_0}$. Individual rationality excludes those objects, while feasibility has already excluded past objects. Hence $\varphi_{i_0}(\tilde{R}) \in O_C$. Thus all $|C|$ members of C receive distinct objects from the $|C|$-element set O_C, and consequently they receive exactly the objects of O_C.

Step 3: exclude $\varphi_{i_0}(\tilde{R}) = o_{i_0}$. If i_0 holds o_{i_0}, then her predecessor i_{m-1} cannot receive its successor o_{i_0} and, by (5), must hold $o_{i_{m-1}}$. Iterating backwards around C pins every member of C at her endowment. But i_0 strictly prefers $o_{i_{m-1}}$ to o_{i_0} at $\tilde{R}$, while i_{m-1} strictly prefers its successor o_{i_0} to $o_{i_{m-1}}$. They would both gain by exchanging assignments, contradicting pair-efficiency.

Step 4: exclude $\varphi_{i_0}(\tilde{R}) = o_{i_r}$ for $2 \leq r \leq m-1$. Suppose i_0 receives o_{i_r}. Agent i_r cannot hold her endowment, so (5) gives her successor $o_{i_{r+1}}$; propagating forward gives successors to $i_r, i_{r+1}, \ldots, i_{m-1}$, with i_{m-1} receiving o_{i_0}. On the other side, the successor of i_{r-1} is o_{i_r}, already held by i_0, so i_{r-1} must hold her endowment. Propagating backwards pins $i_{r-1}, \ldots, i_1$ at their endowments. By the forward order in (4), i_0 strictly prefers $o_{i_{r-1}}$ to her current object o_{i_r}, while i_{r-1} strictly prefers its successor o_{i_r} to $o_{i_{r-1}}$. Exchanging their assignments contradicts pair-efficiency.

Step 5: conclusion. The only remaining possibility is $\varphi_{i_0}(\tilde{R}) = o_{i_1} = c_{i_0}$. Yet (3) gives $\varphi_{i_0}(\hat{R}) = o_{i_0}$, and $o_{i_1} \hat{P}_{i_0} o_{i_0}$. At the true report $\hat{R}_{i_0}$, agent i_0 therefore gains by reporting $\tilde{R}_{i_0}$, which changes only the position of her own endowment. This contradicts *endowment-position strategy-proofness*.

Thus C cannot be tail-isolated at i_0. In particular, it cannot exhaust the active market, so $C \subsetneq A_t$. □

Proof of the positive part of Theorem 2. Suppose $n = 3$ and that an individually-rational, pair-efficient and endowment-position strategy-proof mechanism differs from TTC. Choose a minimal conflict profile exactly as above. The proof of Lemma 2 and Claim 1 use only these three axioms. The same is true of Lemma 4, except for the sorting in Step 1. If the failed cycle has length three, however, its tail block $\{o_{i_2}\}$ is a singleton, so that sorting is vacuous and respecting-improvement is not used.

A missed cycle of length one is impossible by individual rationality. A missed cycle of length two is impossible by pair-efficiency: by Lemma 2, both members keep their endowments although each strictly prefers the other's. Finally, a missed cycle of length three must occur at $t = 1$ and exhaust the market, because after any earlier TTC cycle fewer than three agents remain. It is therefore tail-isolated, and the proof of Lemma 4, with its vacuous sorting step, gives a contradiction. Hence the mechanism is TTC. □

5.2. Separation and induction

We now separate the failed cycle by editing outsiders' reports and use the resulting profile to induce a smaller mechanism inheriting all three axioms. Call a profile Q **C-separated** if

$$o_x \, P_x^Q \, o_i \qquad \text{for every } x \notin C \text{ and } i \in C.$$

Agents removed before stage t already satisfy this separation at the minimal conflict profile by Claim 2(b).

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Lemma 5. *Suppose there is a C-separated profile Q^0 such that*

$$\varphi_i(Q^0) = o_i \qquad (i \in C), \tag{6}$$

and, writing $R_i^C = Q_i^0|_{O_C}$,

$$R_i^C : \quad c_i \succ o_i \succ O_C \setminus \{c_i, o_i\}. \tag{7}$$

Then there is a mechanism on the C-agent housing market satisfying individual-rationality, pair-efficiency, respecting-improvement and endowment-position strategy-proofness, and this mechanism selects no trade at $R^C = (R_i^C)_{i\in C}$ although TTC executes C there.

Proof. Fix the outsiders' reports at Q^0_{-C}. At every profile of the form $(\cdot, Q^0_{-C})$, individual rationality and C-separation prevent each outsider from receiving a cycle object. Feasibility then implies

$$\varphi_x \in O \setminus O_C \ (x \notin C), \qquad \varphi_i \in O_C \ (i \in C). \tag{8}$$

For $i \in C$, let S_i be the order that Q_i^0 induces on $O \setminus O_C$. Starting from Q^0, stably sort Q_i^0, for one $i \in C$ at a time, by moving cycle objects down across outside objects until the report becomes

$$Q_i^B : S_i \succ R_i^C. \tag{9}$$

We show inductively that every member of C keeps her endowment after each adjacent transposition. Suppose the cycle object moved down is o_k. If $k \neq i$, the profile before the move is an improvement for k relative to the profile after it, so Lemma 3 pins k at o_k. If $k = i$, the move is an endowment repositioning. From the inductive hypothesis and *endowment-position strategy-proofness*,

$$o_i = \varphi_i(\text{before}) \ R_i^{\text{before}} \ \varphi_i(\text{after}).$$

By (8) the latter assignment belongs to O_C. The displayed *endowment-position strategy-proofness* inequality together with (7) excludes c_i, which lies above o_i in the unchanged relative order of cycle objects. Individual rationality at the profile after the move excludes every other cycle object, each of which lies below o_i there. Hence i still receives o_i.

In either case one member of C has been pinned. Equation (8), individual rationality and (7) restrict every member $h \in C$ to $\{c_h, o_h\}$. If k holds o_k, her predecessor cannot receive its successor o_k and must hold her own endowment; iterating backwards around C pins the whole cycle. Thus the sorting ends at a profile Q^B satisfying

$$\varphi_i(Q^B) = o_i \qquad (i \in C). \tag{10}$$

Now define the block extension $E_i(L_i) = S_i \succ L_i$ for every linear order L_i of O_C, and put

$$R^E(L) = \big((E_i(L_i))_{i\in C}, Q^0_{-C}\big), \qquad \psi_i(L) = \varphi_i(R^E(L)).$$

Equation (8) makes ψ a mechanism on C and O_C. It inherits individual rationality and pair-efficiency because each extension agrees with the small-market report on O_C. It

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inherits *endowment-position strategy-proofness* because an endowment repositioning of L_i maps to an endowment repositioning of $E_i(L_i)$.

Finally, let $\widetilde{L}$ be an improvement for k relative to L in the small market. The full report of owner k is unchanged. In every other insider's report, only o_k may move upward within the lower O_C block, and the relative order of all objects other than o_k is unchanged; the outside blocks and all outsider reports are fixed. Thus $R^E(\widetilde{L})$ satisfies all three defining conditions for an improvement for k relative to $R^E(L)$ in the full market. Respecting-improvement for φ gives

$$\varphi_k(R^E(\widetilde{L}))\ E_k(L_k)\ \varphi_k(R^E(L)).$$

Both assignments lie in O_C, where $E_k(L_k)$ agrees with the small-market order L_k. Hence $\psi_k(\widetilde{L})$ is weakly preferred to $\psi_k(L)$ according to L_k, so ψ respects improvement.

At R^C, equations (9) and (10) give $R^E(R^C) = Q^B$ and $\psi_i(R^C) = o_i$ for every $i \in C$. Condition (7) makes every i point to c_i, so TTC on the small market executes C. Therefore ψ is not TTC. □

To construct the separated base, rule out assignments of past objects. Respecting-improvement carries any such assignment to a common-sink profile; restoring the original TTC prefix then forces a conflict before stage t.

Claim 3. *Let*

$$D_r = C_r(R^*) \quad (r < t), \qquad C = C_t(R^*), \qquad H = A_t \setminus C.$$

The sets $D_1, \dots, D_{t-1}, C, H$ partition N. Suppose that Q differs from R^* only in the reports of agents in H. Fix $i \in C$, and let $\widehat{Q}^i$ be obtained from Q by moving o_i to the top of every Q_h, $h \in H$, while preserving the relative order of all other objects in each such report. Then

$$C_r(\widehat{Q}^i) = C_r(R^*) \qquad (r = 1, \dots, t),$$

with the same within-cycle pointers.

Proof. Put $D_t = C$ and argue by induction on $r = 1, \dots, t$. Suppose that the selector at $\widehat{Q}^i$ has removed $D_1, \dots, D_{r-1}$ with their original pointers and no member of H. Exactly the same original objects have then been removed as before stage r at R^*. For any agent belonging to some D_s, $s \geq r$, her successor in D_s is her best object after $D_1, \dots, D_{s-1}$ have been removed. Every H-object is still available at that later stage, so she ranks the successor above every H-object. Since the successor is already available at stage r, her pointer cannot lead to an H-object. All reports outside H are unchanged.

Meanwhile o_i remains available through stage t, and every member of H points to o_i by construction. Hence no member of H can belong to a cycle: an H-only cycle is impossible because all its members point to $i \notin H$, and a mixed cycle is impossible because no agent outside H points into H. The cycles at stage r are therefore exactly the cycles among the unchanged agents at the corresponding stage of R^*. The fixed selector chooses D_r, with its original pointers, just as it does at R^*. This completes the induction and proves the claim. □

Lemma 6 (Past-object exclusion)**.** *Put*

$$O^- := O \setminus O_t(R^*) = \{o_d : d \notin A_t\}.$$

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If Q differs from R^* only in the reports of agents in $H = A_t \setminus C$, then

$$\varphi_i(Q) \notin O^- \qquad \text{for every } i \in C.$$

Proof. Suppose, to the contrary, that $\varphi_i(Q) = p \in O^-$ for some $i \in C$. Since $Q_i = R_i^*$, individual rationality gives $p\, R_i^*\, o_i$. The object c_i belongs to $O_t(R^*)$, is best in that set, and by Lemma 2 stands immediately above o_i in the full ranking. Because $p \notin O_t(R^*)$, it follows that $p\, P_i^*\, c_i$. Consequently,

$$\{x \in O : x\, R_i^*\, p\} \subseteq O^-. \tag{11}$$

Construct $\widehat{Q}^i$ as in Claim 3. It is an improvement for owner i with respect to Q: agent i's report is unchanged, o_i moves only upward in the reports of the agents in H, and the relative order of all objects other than o_i is unchanged. Respecting-improvement therefore gives

$$\varphi_i(\widehat{Q}^i)\; R_i^*\; \varphi_i(Q) = p.$$

By (11),

$$\varphi_i(\widehat{Q}^i) \in O^-. \tag{12}$$

Claim 3 says that the first $t-1$ TTC cycles at $\widehat{Q}^i$ are exactly $D_1, \ldots, D_{t-1}$. If φ executed all of them, their members would receive exactly all objects in O^-. By feasibility, the member $i \in C$ could then not receive the past object in (12). Hence φ fails to execute some cycle among $D_1, \ldots, D_{t-1}$, so $\widehat{Q}^i$ is a conflict profile and

$$\rho(\widehat{Q}^i) \leq t - 1.$$

Thus $\widehat{Q}^i$ is a conflict profile with $\rho(\widehat{Q}^i) < t$, contradicting the ρ-minimality of R^*. □

Claim 4. *There is a profile Q^0 satisfying the hypotheses of Lemma 5.*

Proof. No report of a member of C will be changed. For each $h \in H = A_t \setminus C$, define a_h once from R_h^* as her best object in $O_t(R^*)$. Treat the agents of H in any fixed order. Before an agent h is treated, her report is still R_h^*. If no object of O_C lies above o_h, make no change. Otherwise Claim 2(a) implies that every such cycle object must equal a_h; strict preferences therefore make it unique. Write it as o_j and move it in h's report to the position immediately below o_h, preserving the relative order of all other objects.

We show inductively that every member of C holds her endowment after each actual edit. The assertion holds initially by Lemma 2. Suppose it holds before an edit that lowers o_j in h's report. The profile before the edit is an improvement for owner j relative to the profile after it, so Lemma 3 gives $\varphi_j = o_j$ after the edit.

The edited profile differs from R^* only in the reports of agents in H. Lemma 6 therefore excludes every object in $O \setminus O_t(R^*)$ from the assignment of each member of C. Individual rationality, the residual optimality of c_i, and Lemma 2 now imply

$$\varphi_i \in \{c_i, o_i\} \qquad (i \in C).$$

Since j holds o_j, her predecessor on C cannot receive its successor o_j and must hold its own endowment. Iterating backwards around the cycle pins every member of C at her endowment. This completes the induction.

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At the terminal profile Q^0, every active outsider ranks her endowment above every cycle object by construction. Every agent removed before stage t does so by Claim 2(b), and none of those reports was changed. Hence Q^0 is C-separated. The induction gives $\varphi_i(Q^0) = o_i$ for all $i \in C$. Finally, because the reports of the members of C were never changed, residual optimality and Lemma 2 give

$$Q_i^0|_{O_C} : \quad c_i \succ o_i \succ O_C \setminus \{c_i, o_i\}.$$

Thus all hypotheses of Lemma 5 hold. □

Proof of Theorem 1. TTC satisfies all three axioms, as noted in Section 4. For uniqueness, proceed by strong induction on n. The case $n = 3$ is the positive part of Theorem 2.

Let $n > 3$, assume the theorem holds for every smaller population of at least three, and suppose that $\varphi \neq \tau$ satisfies pair-efficiency, respecting-improvement and endowment-position strategy-proofness. Pair-efficiency and respecting-improvement supply individual rationality. Choose the minimal conflict profile R^* and let $C = C_t(R^*)$ be its first failed TTC cycle.

A cycle of length one cannot be missed under individual rationality. A cycle of length two cannot be missed: Lemma 2 leaves both members with their endowments although each strictly prefers the other's, contrary to pair-efficiency. Hence $|C| \geq 3$. Lemma 4 gives $C \subsetneq A_t$, and therefore $|C| < n$.

Claim 4 supplies a separated base. Applying Lemma 5 produces a mechanism on the $|C|$-agent housing market that satisfies pair-efficiency, respecting-improvement and endowment-position strategy-proofness but differs from TTC. Relabel the agents of C and their endowments by $\{1, \ldots, |C|\}$; feasibility and all three axioms are invariant under this bijection. Since $3 \leq |C| < n$, the induced mechanism then contradicts the induction hypothesis. Therefore $\varphi = \tau$. □

6. Concluding remarks

Endowment-position strategy-proofness holds fixed an agent's ranking of trade targets and varies only her endowed object's position. Together with pair-efficiency and respecting-improvement, this one-dimensional condition characterizes TTC and thereby recovers full and group strategy-proofness. Both *endowment-position strategy-proofness* and *respecting-improvement* admit exact adjacent tests, so the characterization rests on three locally testable conditions operating on complementary margins.

The result also locates the boundary of that restriction. At the pair-efficiency level, if movements of the endowment are constrained only in the upward direction, the characterization fails with three agents. If both directions are retained but invariance is imposed only at the report that truncates immediately below the assignment, it fails with four. Respecting-improvement is equally substantive: replacing it by individual-rationality preserves the characterization only for three agents and fails for every larger population.

The proof's central propagation is cross-fiber. Pair-efficiency and respecting-improvement first imply individual rationality, while *endowment-position strategy-proofness* gives constant-endowment or threshold behavior within each fiber. At a minimal departure from TTC, respecting-improvement transports any assignment of a past object to a common-sink profile. The restored TTC prefix then creates an earlier

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conflict. Past-object exclusion and downward pinning separate the failed cycle and produce the smaller counterexample ruled out by strong induction.

The characterization is established for deterministic mechanisms on the strict, unrestricted, one-agent–one-object Shapley–Scarf domain. The same incentive-content question extends naturally to weak preferences, multiple endowments and randomization. More broadly, the common-sink argument illustrates how an ownership-response axiom can bridge report classes that a restricted incentive condition controls only internally.

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Appendix A. The nested-cycle mechanisms

This appendix constructs, for every $n \geq 4$, a mechanism satisfying individual-rationality, pair-efficiency and endowment-position strategy-proofness that differs from the TTC mechanism, completing the negative half of Theorem 2.

Proof of the negative part of Theorem 2. The construction.

For $i \in \{2, \dots, n\}$ define the successor

$$\sigma(i) = \begin{cases} i+1, & i < n, \\ 2, & i = n, \end{cases}$$

so that σ cycles the agents $2, \dots, n$. Fix the reduced orders

$$\widehat{R}_1 : \ o_2, \ \dots \qquad \text{and} \qquad \widehat{R}_i : \ o_{\sigma(i)}, \ o_1, \ \dots \quad (i = 2, \dots, n),$$

where the omitted tails are any fixed rankings of the remaining objects. None of the arguments below uses their internal order, so every fixed choice of these tails gives the stated construction. Let $\mathcal{D}_n \subseteq \mathcal{R}^N$ be the set of profiles R with $R_i|_{O\setminus\{o_i\}} = \widehat{R}_i$ for every i. For $R \in \mathcal{D}_n$ write $r_i = r_i(R_i)$ as in Section 3.

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Define φ^n as follows. If $R \notin \mathcal{D}_n$, or if $R \in \mathcal{D}_n$ and $\min_{i \in N} r_i(R_i) = 0$, set $\varphi^n(R) = \tau(R)$. Otherwise $R \in \mathcal{D}_n$ with $r_i \geq 1$ for every i; put

$$m(R) = \max\big(\{i \in \{2, \ldots, n\} : r_i \geq 2\} \cup \{1\}\big)$$

and set

$$\varphi_i^n(R) = \begin{cases} o_{i+1}, & 1 \leq i < m(R), \\ o_1, & i = m(R), \\ o_i, & i > m(R). \end{cases} \tag{A.1}$$

When $m(R) = 1$ this is the endowment allocation; when $m(R) = m \geq 2$ it executes the prefix cycle $1 \to 2 \to \cdots \to m \to 1$ and leaves every agent above m with her endowment. The mechanism thus runs a nested family of cycles of lengths $1, 2, \ldots, n$. The selected length is the largest index among the agents $2, \ldots, n$ who have declared more than one object acceptable, and is 1 if there is no such agent.

It differs from TTC.

Take $r_i = 1$ for every i. Agent 1's top object is then o_2 and, for $i \geq 2$, agent i's top object is $o_{\sigma(i)}$, so the first-choice graph contains the cycle $2 \to 3 \to \cdots \to n \to 2$ of length $n - 1 \geq 3$. The TTC algorithm executes it, and agent 1 receives o_1 in the following round. But $m(R) = 1$, so $\varphi^n(R)$ is the endowment allocation. Hence $\varphi^n \neq \tau$. This is exactly the configuration pair-efficiency cannot see: the unexecuted improvement cycle has length $n - 1 \geq 3$.

Individual-rationality.

Outside the patched region $\varphi^n = \tau$. Inside it, if $m = 1$ every agent keeps her endowment. If $m \geq 2$, each $i < m$ receives o_{i+1}, the first object of $\widehat{R}_i$, which is above o_i because $r_i \geq 1$; agent m receives o_1, the second object of $\widehat{R}_m$, which is above o_m because the definition of m gives $r_m \geq 2$; and each $i > m$ keeps o_i.

Pair-efficiency.

Suppose first $m = 1$, so $r_i = 1$ for every $i \geq 2$. Then the only object agent $i \geq 2$ prefers to her endowment is $o_{\sigma(i)}$, so the improvement arrows among $2, \ldots, n$ form the single cycle $(2, 3, \ldots, n)$, of length $n - 1 \geq 3$; there is no two-cycle among them. Agent 1 may prefer several objects to o_1, but every $i \geq 2$ has $r_i = 1$ and therefore ranks o_1 below o_i, so no agent would reciprocate an exchange with her.

Now suppose $m \geq 2$. Agents $1, \ldots, m - 1$ receive their top objects and so belong to no improving pair. Agent m receives o_1, and the only object she prefers to it is $o_{\sigma(m)}$. If $m < n$ that object is o_{m+1}, held by agent $m + 1$, who has $r_{m+1} = 1$ by maximality of m and hence ranks o_1 below her own endowment; if $m = n$ it is o_2, held by agent 1, who has her top object. Every $j > m$ keeps o_j, has $r_j = 1$, and wants only $o_{\sigma(j)}$. If $j < n - 1$, that object is held by agent $j + 1$, who wants only o_{j+2} and hence does not want o_j. If $j = n - 1$, it is held by agent n, who wants only o_2 and hence does not want o_{n-1}. If $j = n$, the desired object is o_2, held by agent 1, who has her top object. Thus no two agents want each other's assignments.

Endowment-position strategy-proofness.

An endowment repositioning preserves every agent's reduced order, so it cannot move a profile into or out of $\mathcal{D}_n$: every endowment-position fiber lies entirely inside $\mathcal{D}_n$ or entirely outside it. Outside, and on any fiber inside $\mathcal{D}_n$ along which some other

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agent has $r_j = 0$, the mechanism coincides with τ, which is strategy-proof. It remains to inspect the fibers inside $\mathcal{D}_n$ on which every r_j, $j \neq i$, is at least one. By Lemma 1 it suffices to check that the assignment to the varying agent is constant or of threshold form. Note that at $r_i = 0$ the agent ranks o_i first, the profile lies in the unpatched region, and τ returns o_i.

For agent 1 the value of m does not depend on r_1. If $m = 1$ she receives o_1 at every position; if $m \geq 2$ she receives o_1 at $r_1 = 0$ and o_2 at every $r_1 \geq 1$, a threshold at the first object of $\widehat{R}_1$.

For $i \geq 2$, suppose first that some $j > i$ has $r_j \geq 2$. Then $m \geq j > i$ whatever r_i is, so agent i receives o_i at $r_i = 0$ and $o_{\sigma(i)}$ at every $r_i \geq 1$: a threshold at the first object of $\widehat{R}_i$. Otherwise no $j > i$ has $r_j \geq 2$. Then $m = i$ exactly when $r_i \geq 2$, in which case agent i receives o_1; and when $r_i \leq 1$ we have $m < i$, so agent i keeps o_i. The assignment is thus o_i at $r_i \in \{0, 1\}$ and o_1 at every $r_i \geq 2$: a threshold at the second object of $\widehat{R}_i$.

Every fiber has the form permitted by Lemma 1, so φ^n is endowment-position strategy-proof. □

Remark 3. At $r = (0, 1, \ldots, 1)$, raising o_2 in agent 1's report drops agent 2 from o_3 to o_2, violating respecting-improvement. It also violates upper invariance. For $n = 4$, consider

$$R_1 : o_2, o_1, o_3, o_4, \quad R_2 : o_3, o_1, o_2, o_4,$$
$$R_3 : o_4, o_1, o_2, o_3, \quad R_4 : o_2, o_1, o_3, o_4.$$

Here $\varphi_1^4(R) = o_2$. Swapping only o_3 and o_4, both below o_2 in R_1, moves the profile outside $\mathcal{D}_4$, where TTC assigns o_1 to agent 1.

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